ARYAN GOSWAMI

Software Engineering (C++, Python, Backend Systems) • Applied AI & Machine Learning • 2027 CSE Graduate
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EDUCATION

Atal Bihari Vajpayee Government Institute of Engineering and Technology, Shimla **2023 – 2027 (Expected)**
B.Tech in Computer Science and Engineering • **CGPA: 7.86 / 10.0** (through 6th Semester) **Shimla, Himachal Pradesh**
Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP (C++), Linear Algebra, Probability & AI/ML.

TECHNICAL SKILLS

Languages & Core CS:	**C++ (STL, DSA)**, **Python**, **SQL**, TypeScript, JavaScript; OOP, OS Concurrency, REST Architecture
Backend & Databases:	**FastAPI**, Next.js, Node.js, Pydantic, Prisma ORM, **PostgreSQL**, SQLite, Relational Indexing
Applied AI & ML:	LLM Tool Calling, Agentic Workflows, Structured Outputs, Prompt Engineering, Context Management, Scikit-Learn, NumPy
ML Evaluation & Security:	Adversarial Robustness (**FGSM, PGD, Label-Flip Poisoning**, Decision-Only Black-Box Search), Train/Test Split, Precision, Recall, F1, MAE, R^2
DevOps & Testing Tools:	**Docker** (Sandboxed Worker Isolation), Git, GitHub Actions (CI/CD), **pytest**, Groq API, Gemini API, Linux CLI

EXPERIENCE

AI & Software Engineering Intern • *Infosys Springboard 7.0* | Code • Live Demo **Jul 2026 – Sep 2026 | Remote**

- **Architected and deployed** a full-stack smart facility operations platform using **Python, FastAPI, Groq LLM APIs, and PostgreSQL** to automate telemetry analysis, work-order generation, and real-time alerting across energy, maintenance, occupancy, security, and cost.
- **Engineered data ingestion and evaluation pipelines** integrating **35,040 time-series energy readings** and model predictions into operational dashboards, backed by automated `pytest` suites verifying regression error (MAE, R^2) and classification Precision, Recall, and F1.
- **Designed idempotent fallback and error-recovery workflows** with persistent PostgreSQL state tracking so upstream LLM provider errors or interrupted diagnostic runs safely resume without replaying completed database mutations.

KEY ENGINEERING PROJECTS

AI Software Engineer Workspace | *Next.js, TypeScript, Prisma, PostgreSQL, Docker, Groq* | Code • Live Demo **2026**

- **Built an agentic software engineering workspace** that converts natural-language feature requests and saved repository context into multi-file code outputs, structured task lists, and change explanations via low-latency Groq LLM inference.
- **Implemented a strict output-validation layer** that parses structured model responses and rejects malformed, truncated, or duplicate file writes while persisting per-account conversation history and project context in **PostgreSQL via Prisma**.
- **Engineered an isolated Docker execution worker** that compiles and tests generated Node.js code inside a **network-disabled container with a strict 30-second timeout**, fully decoupling untrusted code execution from output validation.

Adversarial ML Red-Teaming Platform (Team Lead, Team of 4) | *Python, NumPy, FastAPI, Docker* | Code • Demo **2026**

- **Led end-to-end software engineering for a 4-person team**, owning the FastAPI service architecture, authentication, artifact storage, automated vulnerability reporting, and containerized Docker deployment alongside ML evaluation.
- **Implemented white-box, black-box, and poisoning attack evaluators** on linear-softmax classifiers using **FGSM, PGD, decision-only random search, and label-flip poisoning** to benchmark clean accuracy, adversarial accuracy, and attack success rates.
- **Enforced deterministic reproducibility and leakage prevention** through automated unit tests checking L_∞ perturbation bounds, repeatable PGD trajectories, clean-retraining defense baselines, and strict train/test dataset separation.

Yatra Kavach — Offline-First Safety Platform (Backend Lead, Team of 4) | *REST APIs, Offline Sync* | Live Demo **2026**

- **Led backend development for a 4-member hackathon team** building an offline-first route-safety and emergency SOS platform designed for high-altitude Himachal Pradesh pilgrimage corridors with unreliable cellular connectivity.
- **Engineered resilient local-to-cloud state synchronization** and compact telemetry payloads so emergency alerts, last-known GPS coordinates, and route checkpoints queue offline and synchronize atomically upon network recovery.

ACHIEVEMENTS & CERTIFICATIONS

- **Competitive Programming & DSA:** Solved **300+ algorithmic problems on** LeetCode in C++ and Python spanning Arrays, Binary Search, Trees, Graphs, Heaps, and Dynamic Programming.
- **Hackathon Awards: 1st Place Winner**, ABVGIET Ideathon (Nov 2025) • **Finalist**, SLSH State-Level Hackathon (Oct 2026) • **2nd Place**, Smart India Hackathon (SIH) College Round (Aug 2026).
- **Certifications: Oracle Cloud Infrastructure 2025 Certified Generative AI Professional • Oracle Certified Foundations Associate — Agentic AI (2026) • Infosys Springboard (2026):** Principles of Generative AI & Prompt Engineering.